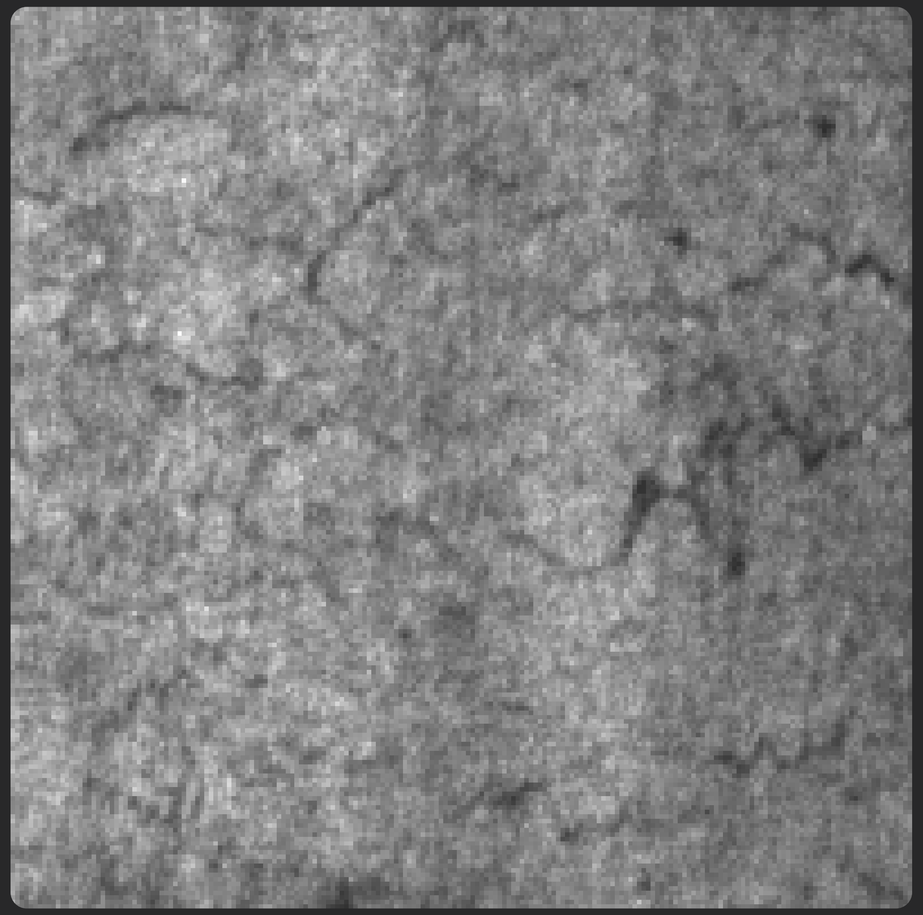
Classifying steel surface defects

Given a cropped steel image, predict one of six NEU-CLS defect classes.

Baseline: ResNet-18

ImageNet pretrained · minimal training augmentation

Fixed split · 1,259 train / 180 validation / 360 test



Cr_107 / Ground truth: Crazing

Camera-aware training raised F1

ResNet-18 vs. EfficientNet-B0 on clean crops and 27 synthetic corruptions.

MODEL / TRAINING	CORRUPTED MACRO-F1	CLEAN MACRO-F1
ResNet-18 Minimal · Baseline	**0.8293 ± 0.0077**	1.0000 ± 0.0000
ResNet-18 Camera-aware	**0.9577 ± 0.0152**	0.9963 ± 0.0032
EfficientNet-B0 Minimal	**0.7387 ± 0.0346**	0.9963 ± 0.0016
EfficientNet-B0 Camera-aware	**0.9501 ± 0.0139**	0.9954 ± 0.0058

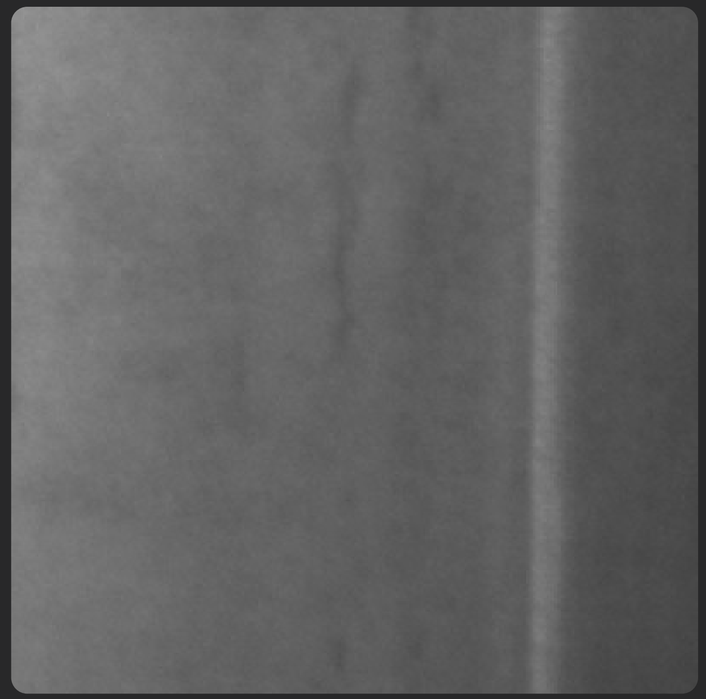


Mean ± sample SD across 3 seeds. Bars use a 0–1 scale.

Corrupted F1 improved for both models. Clean F1 stayed near 1.00.

A clean scratch was misclassified

Sc_69 is an unmodified crop from the held-out test set.



GROUND TRUTH → TOP-1 PREDICTION

Scratches → Inclusion

EfficientNet-B0 minimal · seed 43

RECORDED TOP-2 PROBABILITIES

55.3% inclusion / 39.6% scratches

The faint, elongated mark may resemble an inclusion.

Live demo: clean Cr_107 → same crop with Gaussian blur → Sc_69.

Four pretrained checkpoints from the same training seed (43).